Project Proposal

Level 04

SWOT Tank Water Tracker: Multi-Sensor Deep Learning and Satellite Altimetry for Water Storage Estimation and Forecasting in Sri Lankan Tanks

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1. Introduction

Agriculture in the dry zone of Sri Lanka depends on irrigation tanks, from large reservoirs to small village tanks linked in cascades, and FAO has designated this tank-village system a Globally Important Agricultural Heritage System [2]. Storage records, however, exist only at the top of that hierarchy: the Irrigation Department publishes daily figures for its major and medium reservoirs alone [8]. Thousands of village tanks have neither a gauge nor a storage record, so each Yala and Maha season is planned without any measure of their storage.

This project proposes to estimate that missing storage from free satellite data. The Surface Water and Ocean Topography (SWOT) satellite of NASA and CNES measures the water surface elevation of lakes as small as 250 m × 250 m [12], while Sentinel-1 radar and Sentinel-2 optical imagery record their surface area. The proposed system combines the two through a deep-learning water map, a storage model that converts height and area into fractional storage (% full) and a model that forecasts storage 2 to 4 weeks ahead.

2. Background & Motivation

Satellite mapping of tank water is mature; what it does not supply is volume. IFAD and ESA produced monthly 10 m water-surface maps for over 10,650 water bodies in six districts [7], yet tanks of equal area can hold very different volumes. Perera and Jayaweera related Sentinel-1 water area to reported volumes (R^2 of 0.67 to 0.94) [13], and Madumalsha et al. matched ICESat-2 heights to gauge levels with a correlation of about 0.99, although the laser revisits each site only every 91 days [10]. Meyen's water-level forecasts at the University of Moratuwa relied on gauge records [11], which village tanks lack.

SWOT removes both limitations: it measures water height without a gauge and observes each study tank about twice every 21 days. Ma et al. estimated storage from SWOT height and area within 10% at twelve western US reservoirs [9] and Das and Hossain tracked storage change at 245 reservoirs in five regions [3]. Two problems remain. The first is concealment: monsoon cloud blocks optical imagery [13] and floating vegetation such as water hyacinth covers the surface [4]; Pruekthikanee's 2026 abstract separates open water from floating plants with a U-Net, but without SWOT [14]. The second is reliability: SWOT flags almost every small-tank observation as "suspect". Fjørtoft and Pottier report preliminary errors of 12 to 15 cm for lakes between 250 m × 250 m and 1 km² [5], yet in this project's tests at eighty-four Sri Lankan reservoirs, retaining suspect observations raised the mean absolute error from 12 to 21 cm. Free satellite data could therefore supply storage estimates for unmonitored tanks, and the preliminary tests in Section 5.4 indicate that this is feasible.

3. Problem in Brief: No Storage Figures for Village Tanks

Village tanks have no gauges or storage records, and satellite water maps show only their area. The specific problem is to estimate, from free satellite data, how full each tank is and how its storage will change, even when clouds or floating plants hide the water. The project has four research questions:

RQ1. How accurately can deep learning map tank water from Sentinel-1 and Sentinel-2, including tanks under cloud or partly covered by water plants?

RQ2. How accurately does SWOT measure water height on major reservoirs and on village tanks close to its 6.25 ha size limit, compared with gauge readings, and how many "suspect" readings can a learned screen keep, and at what error?

RQ3. Can a model trained on about 100 gauged reservoirs turn a tank's SWOT height and AI water area into % full, for tanks it has never seen, more accurately than a single area-storage formula?

RQ4. Can a deep-learning model forecast each tank's storage 2 to 4 weeks ahead, from recent storage, rainfall and evaporation, better than a "no change" forecast?

4. Aim & Objectives

Aim: The aim of this project is to develop a system for estimating and forecasting the water stored in Sri Lankan tanks, including village tanks without gauges, with the use of SWOT satellite altimetry, Sentinel-1 and Sentinel-2 images and machine learning.

Objectives:

1. Study tank storage monitoring in Sri Lanka and the satellite and machine-learning methods used elsewhere.
2. Build a labelled dataset of Sri Lankan tank water masks, including cloudy and plant-covered cases, and train a deep-learning water map (RQ1).
3. Process SWOT heights for the study tanks, check them against gauge readings and train a screen that decides which readings to keep (RQ2).
4. Train a storage model on about 100 gauged reservoirs that turns SWOT height and water area into % full for any tank (RQ3).
5. Train a deep-learning model to forecast the storage of each tank 2 to 4 weeks ahead (RQ4).
6. Design and develop an integrated system with a web dashboard for the 13 study tanks.
7. Evaluate the proposed solution against gauge records and the Kanhindawewa drone survey.
8. Prepare the final documentation.

5. Proposed Solution: A Satellite and Machine-Learning Storage Tracker

5.1 Study Area and Data

The study area comprises 13 tanks east of Anuradhapura: five major reservoirs with daily gauge records (Hurulu Wewa, Mahakanadarawa, Manankattiya, Nachchaduwa and Nuwara Wewa) and eight village tanks within about 5 km of Mahakanadarawa, all included in the SWOT Prior Lake Database. Table 1 summarises the data sources.

Table 1: Data sources and their use (A, B and C denote the modules)

Data	Source	Use
SWOT Lake Single-Pass product	NASA and CNES, via the Hydrocron API	Water surface elevation (B)
Sentinel-1 SAR and Sentinel-2 imagery	Copernicus, via Google Earth Engine	Water mapping (A)
Sen1Floods11 and a subset of S1S2-Water	Labelled training datasets [1], [15]	Water-map pre-training (A)
Daily storage of 117 reservoirs	Irrigation Department [8], archived daily	Training and validation (B, C)
Water surface area of about 70 reservoirs	Global Water Watch archive [6]	Storage model training (B)
Rainfall and evaporation	GPM IMERG and CHIRPS; ERA5	Forecast inputs (C)
Kanhindawewa drone (LiDAR) survey	Irrigation Department, 2026	Validation (area and volume)
Village-tank gauge readings, if available	Field observations on SWOT pass dates	SWOT height validation (B)

5.2 System Overview

As Figure 1 shows, satellite imagery, SWOT water surface elevations, daily storage records and meteorological data pass through a chain of models in three modules. The fractional storage, storage change and storage forecast of each tank are presented on a web dashboard for irrigation officers and farmer organisations and updated after each new satellite image or SWOT pass.

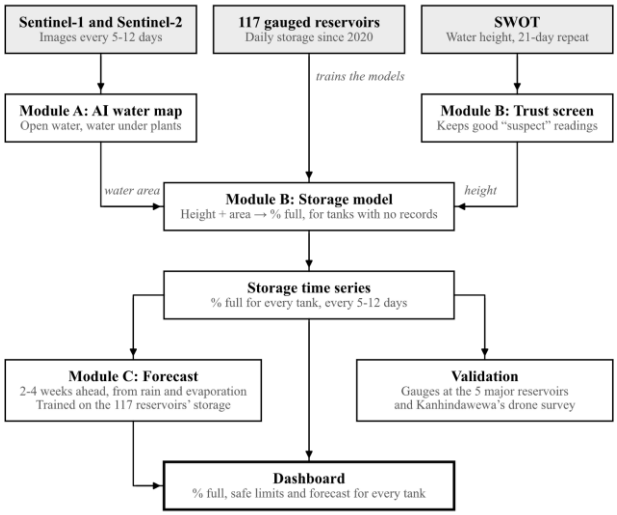


Figure 1: The proposed system (A, B and C are the three members' modules)

5.3 Modules and Individual Scopes

Module A: AI water map (RQ1). A U-Net segmentation network is pre-trained on Sen1Floods11 and S1S2-Water, then fine-tuned on our own labelled masks with three classes: open water, water under plants and land. Because it reads radar and optical images together, it can still map water under cloud. It is compared with water-index and radar-threshold methods using IoU and F1, separately for cloudy and plant-covered tanks.

Module B: SWOT heights and storage (RQ2, RQ3). SWOT heights are checked against gauges at the five major reservoirs. A trust screen (gradient boosting) scores each reading from SWOT's own quality fields and decides which to keep. A storage model (LightGBM) then estimates fractional storage (% full) from SWOT height and water area, with outline and catchment attributes from the Copernicus elevation model; it is trained on about 100 gauged reservoirs and evaluated on reservoirs withheld from training. A height-area-volume relationship for each study tank provides storage change and an independent check on the model.

Module C: Forecast and system (RQ4). An LSTM network forecasts storage 2 and 4 weeks ahead from recent storage, seasonality, rainfall and evaporation. Trained on the daily records of 117 reservoirs, it is compared with gradient boosting and a "no change" forecast by mean absolute error on years and reservoirs withheld from training. Module C also integrates the three modules into an automated pipeline and develops the web dashboard.

5.4 Feasibility and Evaluation

All 13 tanks are included in the SWOT lake database, with 47 to 90 usable observations each between July 2023 and October 2026. At the five major reservoirs, SWOT heights lie 0.40 to 0.64 m above levels derived from the department's storage records; once this constant offset is removed, the scatter is only 0.07 to 0.13 m. On Irrigation Department data, keeping the same 85% of observations as SWOT's "good or suspect" setting, the trust screen reduced the mean absolute error from 21 to 15 cm. On reservoirs withheld from training, a height-only storage model erred by about 8% of capacity, against 11% for a single height-storage formula (11% against 14% under 2 km²). Relative to a "no change" forecast, gradient boosting reduced the 2 and 4-week errors by 13% and 22%, and the LSTM by 9% and 20%. The principal risk is water concealed beneath floating vegetation; if mapping fails there, the storage model can operate on SWOT height alone.

6. Resource Requirements

- **Hardware and field equipment:** the team members' laptops; smartphones with camera and GPS for recording gauge readings; travel to tanks within about 40 km of Anuradhapura.
- **Cloud computing, data and software:** free cloud GPUs (Google Colab or Kaggle, 16 GB), with paid Colab compute units only if the free quota is insufficient; free Google Earth Engine (non-commercial use) and NASA Earthdata accounts; Python with PyTorch, LightGBM, xarray, rasterio and geopandas, QGIS, FastAPI, Next.js, Leaflet and GitHub.
- **Resources already obtained:** a GitHub logger that has archived the Irrigation Department's reservoir data three times daily since 26 October 2026, the project website and a shared 5 TB cloud drive.

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Appendix A: Plan of Action

Table 2 lists the activities, the objectives they serve and their intended dates, from October 2026 to June 2027.

Table 2: Plan of action

No.	Activity	Objective	Module	Start	End
1	Proposal, ethics and proposal presentation	1	All	19 Oct 2026	30 Nov 2026
2	Field check of tank names and gauge posts	1, 3	All	16 Nov 2026	15 Dec 2026
	Milestone: study area confirmed				15 Dec 2026
3	Labelled water masks	2	A (all help)	1 Dec 2026	28 Feb 2027
4	Gauge readings on SWOT pass dates	3, 7	All	4 Jan 2027	30 Apr 2027
5	AI water mapping model	2	A	4 Jan 2027	31 Mar 2027
6	SWOT heights, trust screen and storage model	3, 4	B	18 Jan 2027	15 Apr 2027
7	Storage forecast model	5	C	18 Jan 2027	30 Apr 2027
8	Storage time series and validation	7	B, C	15 Mar 2027	15 May 2027
9	Integrated system and dashboard	6	C	1 Apr 2027	31 May 2027
10	Thesis and research paper	8	All	1 May 2027	30 Jun 2027
	Milestone: final submission		All		30 Jun 2027